TRIP ANALYSIS DASHBOARD

Professional Power BI Project Documentation

Prepared for operational performance, delivery reliability, vehicle utilization and cost analysis

1. Project Overview

The Trip Analysis Dashboard is an interactive business intelligence solution developed in Microsoft Power BI. It converts transportation and delivery data into actionable insights related to delivery performance, travel distance, fuel consumption, vehicle type, driver performance and operating cost.

The dashboard is designed for managers, logistics coordinators and operations teams who need to monitor service quality, identify cost drivers and investigate late deliveries.

2. Objectives

• Measure on-time and late delivery performance.

• Monitor average delivery time and travel distance.

• Calculate fuel-related operating cost and cost per kilometre.

• Compare vehicle types and identify performance differences.

• Identify factors influencing delivery status.

• Support data-driven operational decisions through interactive visuals.

3. Data Model

The report uses the following principal tables identified in the Power BI model:

Table	Purpose
TRIPDB	Trip, delivery, distance, fuel, driver and delivery-status information.
VEHICLEDB	Vehicle master data, including vehicle type.
Local Date Table	Date-based analysis for trends and time filtering.

4. Dashboard Components

Visual	Business purpose
On-Time Delivery % card	Displays the percentage of deliveries completed on time.
Average Delivery Time card	Shows the average time required to complete deliveries.
Average Cost per km card	Shows average operating cost for each kilometre travelled.
Vehicle Type chart	Compares the number or distribution of vehicles by type.
Delivery trend chart	Shows delivery performance across dates or time periods.
Key Influencers	Identifies fields associated with delivery status, such as distance, vehicle type and driver ID.
Decomposition Tree	Breaks down cost per kilometre by vehicle type, maintenance, distance and related dimensions.

5. Key Measures and Calculations

On-Time Delivery %
Percentage of records whose delivery status is classified as On Time.

Average Cost per km
Average of the calculated cost-per-kilometre value across valid trip records.

Cost per km
Fuel consumed × fuel price ÷ distance travelled. A fuel price of ■100 per litre may be used when that is the approved business assumption.

Average Delivery Time
Average duration between the relevant trip start and delivery completion timestamps.

6. Data Preparation and Quality Checks

• Standardize delivery-status values such as On Time and Late.

• Confirm distance and fuel-consumption fields are numeric.

• Replace invalid, blank or negative distance values according to business rules.

• Check vehicle-type spelling and maintain consistent categories.

• Validate date and time columns before creating time intelligence.

• Review missing driver IDs, vehicle IDs and maintenance values.

• Confirm that fuel price assumptions are documented and approved.

7. Dashboard Usage Guide

1. Select a date or period using the date slicer. 2. Review the KPI cards for overall performance. 3. Use the vehicle-type visual to compare categories. 4. Select a delivery-status value in Key Influencers to investigate contributing fields. 5. Expand the Decomposition Tree to analyse cost per kilometre by vehicle type, maintenance and distance. 6. Clear selections to return to the overall view.

8. Business Insights Supported

• Which vehicle types are associated with higher operating cost?

• Does longer distance correspond to increased cost per kilometre?

• Which drivers or vehicle categories show a higher proportion of late deliveries?

• How does delivery performance change over time?

• Where can maintenance or route planning be reviewed to reduce cost?

9. Limitations and Recommendations

Dashboard results depend on the completeness and accuracy of the source data. Cost calculations should be interpreted alongside maintenance, tolls, labour, depreciation and other costs when those fields are available. Before production deployment, establish a refresh schedule, validate relationships, document measure definitions and apply role-based access if driver or customer data is sensitive.

10. Conclusion

The Trip Analysis Dashboard provides a consolidated view of delivery reliability, travel efficiency and operational cost. Its combination of KPI cards, trend analysis, Key Influencers and Decomposition Tree enables users to move from high-level monitoring to detailed investigation and informed operational improvement.